Module

Module Unit: - Gastro Intestinal Tract

Module Title: - liver dialysis

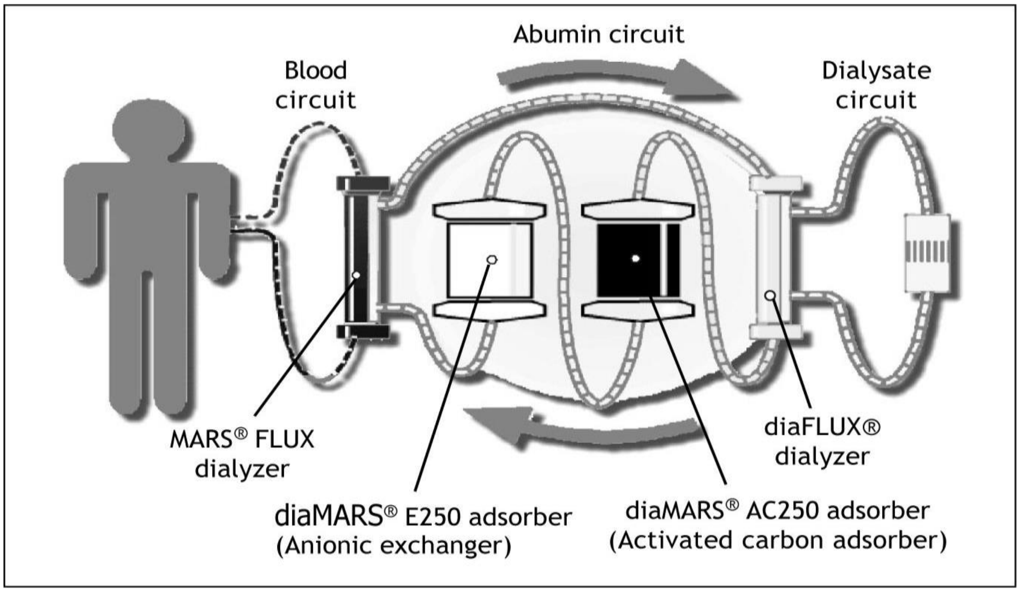
Liver dialysis : is a temporary, artificial, extracorporeal detoxification treatment for severe liver failure that cleans the blood of toxic, protein-bound substances (such as bile acids and ammonia) when the liver can no longer do so. It acts as a "bridge" to support the patient, allowing the liver to recover until a transplant can occur. It operates as a continuous therapy in the ICU for 6–8 hours, using a specialized machine to filter blood through an albumin-rich dialysate, which is then regenerated, allowing the body to recover while awaiting a transplant.

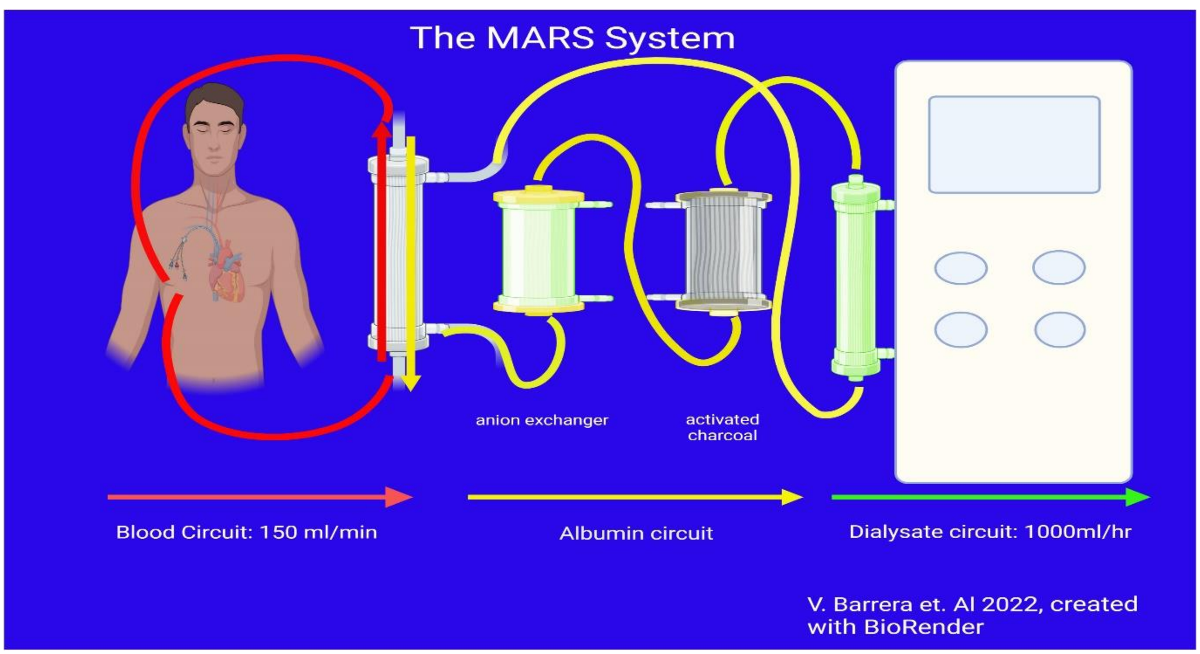
Unlike kidney dialysis which removes water and small solutes, Liver Dialysis must remove protein-bound toxins (like bilirubin and bile acids) and ammonia.

- **Purpose**:
- To support the liver by "cleaning" the blood to improve neurological status (Hepatic Encephalopathy) and allow the liver time to regenerate or bridge the patient to a transplant.
- **Types of Liver Dialysis**

1- The Molecular Adsorbent Recirculating System (MARS): the best extracorporeal liver dialysis system in the medical world. Two separate dialysis circuits are used in the MARS blood purification system. MARS is a widely used liver dialysis technique but rarely used in developing countries due to its complicated technique and high cost.

2- <u>Single Pass Albumin Dialysis (SPAD)</u>: a simple albumin dialysis technique as it uses standard renal replacement therapy machines without the interruption of additional perfusion pump systems like MARS.





Nursing Action	Rationale
Pre-Procedure **Assessment**	
Neurological Baseline: use the Glasgow Coma Scale (GCS).	Liver failure causes ammonia buildup, leading to confusion or coma
Coagulation Profile: check INR, PT, and platelets.	Patients with liver failure have a high risk of bleeding
Vascular Access: Ensure a large-bore double-lumen catheter (usually internal jugular) is patent.	Observe for any inflammation signs as swelling, tenderness.
Prepare and check the main circuits (blood or patient, albumin, and dialysis) and explain the procedure to the patient if conscious or to his family	
During session	
Continuous Monitoring for • Blood pressure (hypotension is common) • Heart rate • Oxygen saturation • Level of consciousness **Observe for complications as** bleeding: • Catheter site, gums , nose • Bruising **Machine-related problems:** • Air bubbles • Clotting • Alarms	

Monitor anticoagulation therapy and document all medications given during dialysis.	
Post- dialysis nursing care	
Reassess: • Vital signs (specially, blood pressure and temperature) • Mental status improvement **Inspecting catheter site** **Follow up post-dialysis lab results** **Documentation:** • Duration of session • Complications • Patient response	Liver dialysis can cause heat loss, leading to shivering, which increases metabolic demand on an already stressed liver.

<u>**Module**</u>

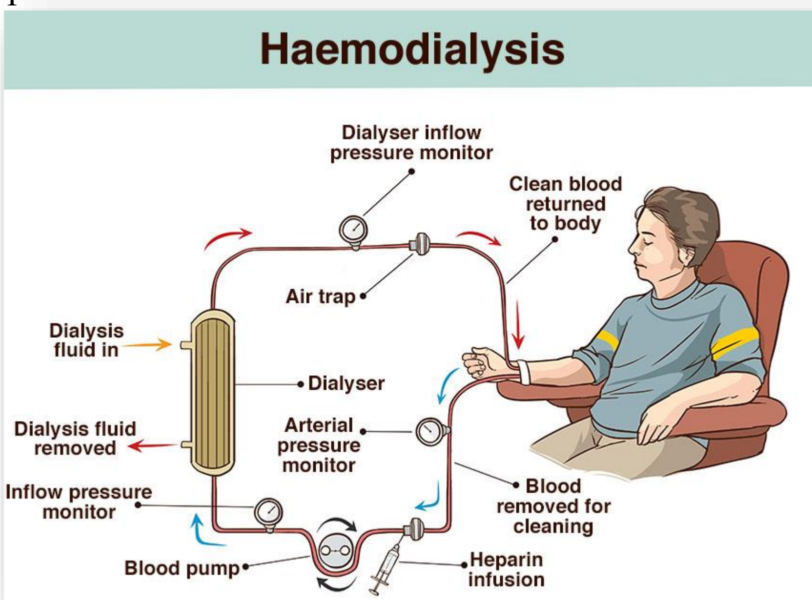
Module Unit: - Renal Procedures

Module Title: - Renal dialysis

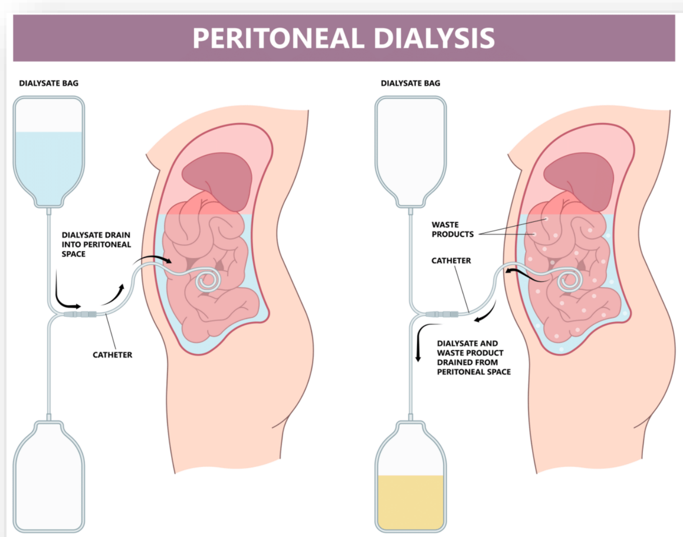
<u>**Definition**</u>: a life-sustaining treatment for end-stage kidney failure that removes waste, toxins, and excess fluid from the blood when kidneys can no longer function

Types of Dialysis:

- <u>**Hemodialysis**</u>**:** Blood is pumped out of the body, filtered through a dialyzer (artificial kidney) machine, and returned. Usually done 3 times a week for 3–5 hours per session.



- <u>**Peritoneal Dialysis**</u>**:** A cleaning solution (dialysate) is instilled into the abdomen via a catheter, allowing waste to be filtered directly inside the body.



Indications of hemodialysis:

- Acute kidney injury
- Uremic encephalopathy
- Pericarditis
- Life-threatening hyperkalemia
- Hypervolemia causing end-organ complications (e.g., pulmonary edema)

Contraindications of hemodialysis:

Absolute contraindication to hemodialysis is the inability to secure vascular access, and relative contraindications include:

- Difficult vascular access
- Needle phobia
- Cardiac failure
- Coagulopathy

Complications

The most common complications associated with hemodialysis are:

- Intradialytic hypotension
- Dialysis disequilibrium syndrome: This is more common in patients during or soon after their first treatment. It is a clinical syndrome characterized by neurologic deterioration, restlessness, mental confusion, headache, occasional muscle twitching, and coma.
- Hemolysis
- Air embolism

Nursing action	**Rationale**
Pre hemodialysis session	
Assess vital signs	To establish a "hemodynamic baseline" and ensure the patient is stable enough to tolerate the stress of fluid removal
Auscultate lung sounds	To detect pulmonary congestion and guide fluid removal (ultrafiltration)
Monitor lab investigations as blood urea, creatinine, sodium, potassium, calcium, albumin and hemoglobin.	to evaluate the severity of uremic toxicity and electrolyte imbalances, **albumin** marker for nutritional status and "oncotic pressure , **hemoglobin**

<table>
<tr><td></td><td>to assess for anemia which is common in renal patients.</td></tr>
<tr><td>Check for holding any antihypertensive medications</td><td>To prevent severe intradialytic hypotension</td></tr>
<tr><td>Observe any inflammation signs as:
✓ Localized erythema
✓ Swelling
✓ Tenderness
✓ Pain in the access arm</td><td>To detect early signs of infection or thrombosis to prevent systemic sepsis and preserve the life-sustaining vascular access</td></tr>
<tr><td>Observe site of access for any abnormalities as:
✓ Vascular swelling
✓ Redness around puncture site
✓ Aneurysm</td><td>To ensure the integrity and patency of the vascular access to prevent life-threatening complications and ensure a successful treatment</td></tr>
<tr><td>Prepare hemodialysis machine, dialyzer, dialysate and supplies</td><td></td></tr>
<tr><td colspan="2">During hemodialysis session</td></tr>
<tr><td><u>Following aseptic technique during cannulation procedure as</u>
➢ Perform hand hygiene
➢ Wearing PPE
➢ Appling skin antiseptic solution appropriately
➢ Allow skin antiseptic solution to dry
➢ No contact with fistula/ graft's site (after disinfection)
➢ Perform cannulation aseptically by avoiding touch needle
➢ Cover puncture site with sterile dressing
➢ Insert needles in vascular access
➢ Remove gloves
➢ Perform hand hygiene</td><td></td></tr>
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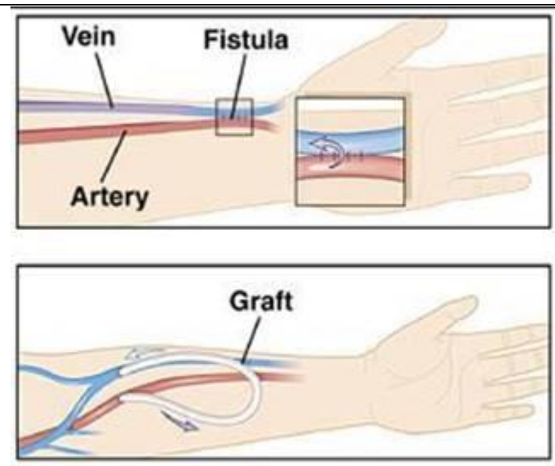
<tr><td></td><td>
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<tr><td>Assess vital signs especially blood pressure every 30 minutes</td><td>To detect and manage intradialytic hypotension early, and ensuring patient safety during rapid fluid shifts</td></tr>
<tr><td>Observe any hemodialysis complications as:
➢ Hypotension
➢ Hypertension
➢ Hypoglycemia
➢ Hyperglycemia
➢ Leg cramps
➢ Febrile reaction
➢ Dysrhythmia
➢ Bleeding at the access site
➢ Monitor patient for disequilibrium syndrome:
✓ Nausea
✓ Vomiting
✓ Restlessness
✓ Headache
✓ Seizures</td><td></td></tr>
<tr><td><u>Observe Vascular access patency</u>
➢ Uncovered during hemodialysis session
➢ Don’t sleep on access arm
➢ Avoid using the patient’s dialysis access arm for administering I.V.</td><td>To ensure continuous blood flow and prevent clotting</td></tr>
</table>

medications, taking BP, or blood sample	
Post- dialysis nursing care	
<u>Following aseptic technique during decannulation procedure as:</u> ➢ Hand hygiene ➢ Wearing PPE ➢ Remove needles aseptically and compress the site of insertion by sterile dressing ➢ Disinfect site of puncture ➢ Cover the site of puncture with sterile gauze or dressing ➢ Removing gloves ➢ Perform hand hygiene	
<u>Assess vital signs</u>	
Auscultate lung sounds	To confirm that the lungs are clear and the fluid removal was successful. And to ensure that patient is leaving the session without any remaining fluid congestion.
Monitor lab investigations as blood urea, creatinine, sodium, potassium, calcium, albumin and hemoglobin.	To confirm the treatment was effective and ensure that waste products and electrolytes have reached safe, stable levels
Instructions given to patients for vascular access maintenance as: ✓ Avoid wearing tight clothes at vascular access arm. ✓ Avoid carrying heavy weight at vascular access arm. ✓ Avoid measuring blood pressure from vascular access arm. ✓ Avoid laying on vascular access arm. ✓ Others	
Discard all equipment and supplies in special places	